



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.10.2025

Mobile Applications for IOS

Introductions to Apple devices and macOS
UI/UX Interactions

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PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30

What is Apple?

Apple is one of the world's leading technology companies, famous for its elegant design, powerful hardware, and smooth user experience.

It creates an ecosystem where all devices – Mac, iPhone, iPad, Apple Watch, and more – work perfectly together.



Mac



iPhone



iPad



Apple Watch



Apple Vision Pro



AirPods



AirTag



Apple TV 4K



HomePod

Mac

Apple's line of computers combines power and design, featuring macOS and Apple Silicon chips that deliver exceptional performance and energy efficiency for creative and professional work.



MacBook Air



MacBook Pro



iMac



Mac mini



Mac Studio



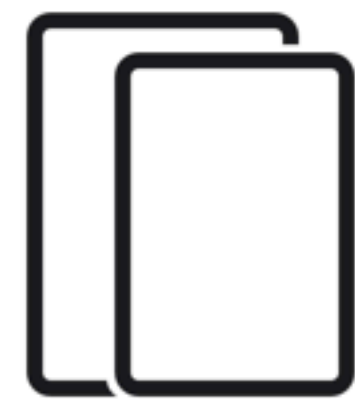
Mac Pro

iPad

The iPad bridges portability and productivity with its intuitive touch interface, Apple Pencil support, and multitasking capabilities – ideal for both creativity and education.



iPad Pro



iPad Air



iPad



iPad mini

iPhone

The iPhone revolutionized mobile technology, offering cutting-edge performance, advanced cameras, and seamless integration with the Apple ecosystem.



iPhone 17 Pro
New



iPhone Air
New



iPhone 17
New



iPhone 16



iPhone 16e

Watch

Designed for health, fitness, and connectivity, Apple Watch tracks activity, monitors wellness, and keeps you in touch – right from your wrist.



Apple Watch
Series 11
New



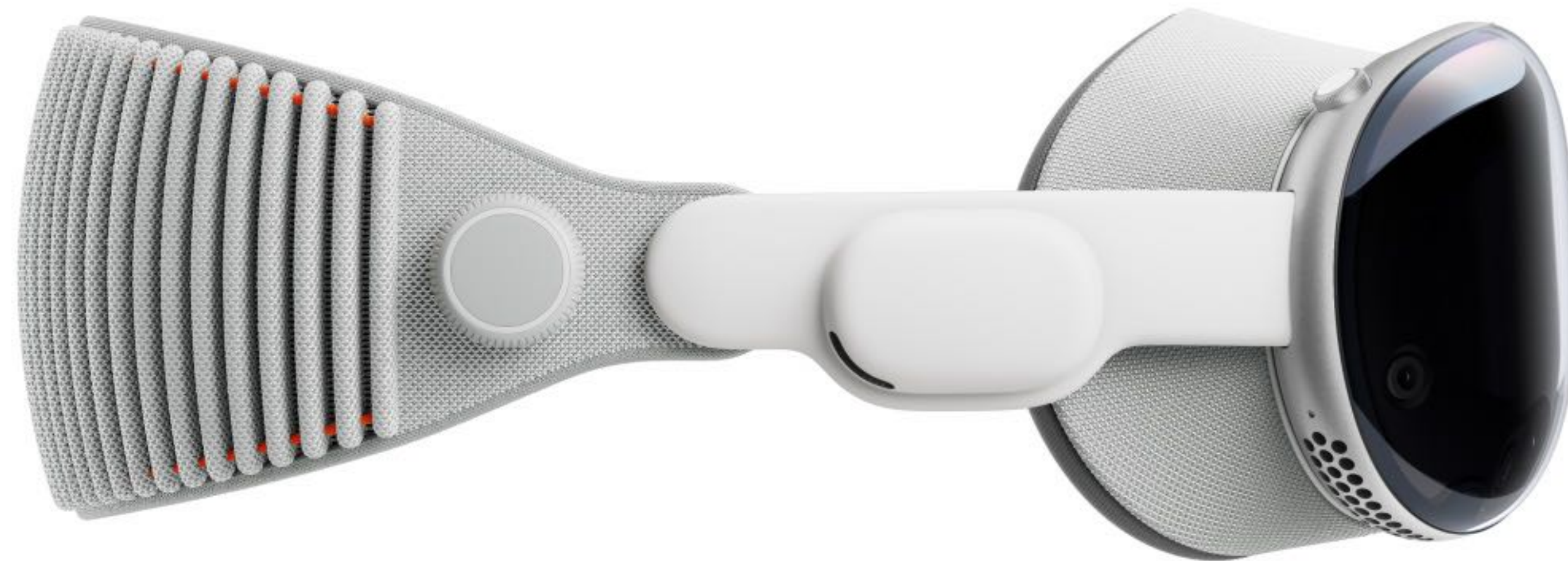
Apple Watch
SE 3
New



Apple Watch
Ultra 3
New

Apple Vision Pro

Apple Vision introduces spatial computing, merging digital content with the real world for immersive experiences in productivity, entertainment, and communication.



Why are we talking about this?

iPhone models compatible with iOS 26 (1)

[iPhone 11](#)

[iPhone 11 Pro](#)

[iPhone 11 Pro Max](#)

[iPhone 12 mini](#)

[iPhone 12](#)

[iPhone 12 Pro](#)

[iPhone 12 Pro Max](#)

[iPhone 13 mini](#)

[iPhone 13](#)

[iPhone 13 Pro](#)

[iPhone 13 Pro Max](#)

[iPhone 14](#)

[iPhone 14 Plus](#)

[iPhone 14 Pro](#)

[iPhone 14 Pro Max](#)

[iPhone 15](#)

[iPhone 15 Plus](#)

[iPhone 15 Pro](#)

[iPhone 15 Pro Max](#)

[iPhone 16](#)

[iPhone 16 Plus](#)

[iPhone 16 Pro](#)

[iPhone 16 Pro Max](#)

[iPhone 16e](#)

[iPhone 17](#)

[iPhone 17 Pro](#)

[iPhone 17 Pro Max](#)

[iPhone Air](#)

iPhone models compatible with iOS 26 (2)

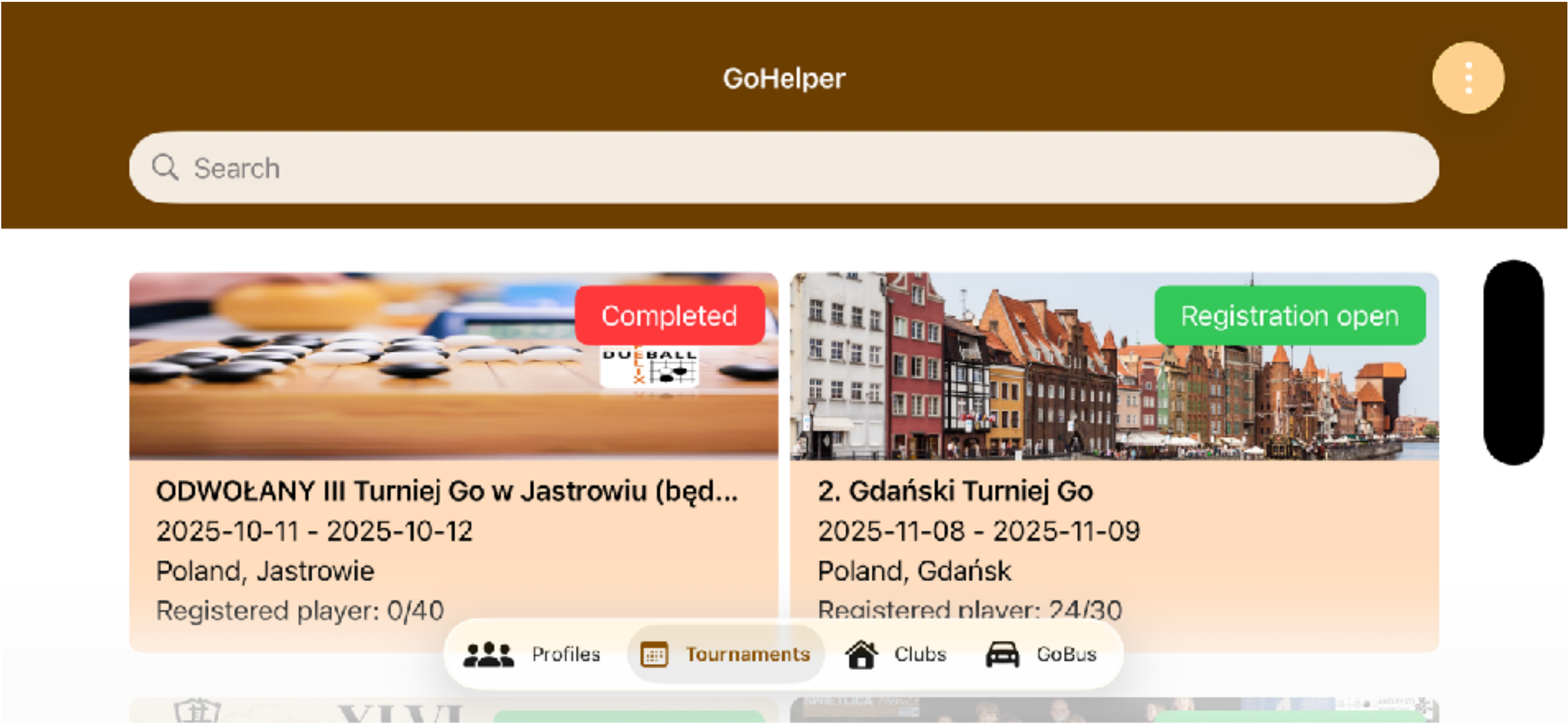
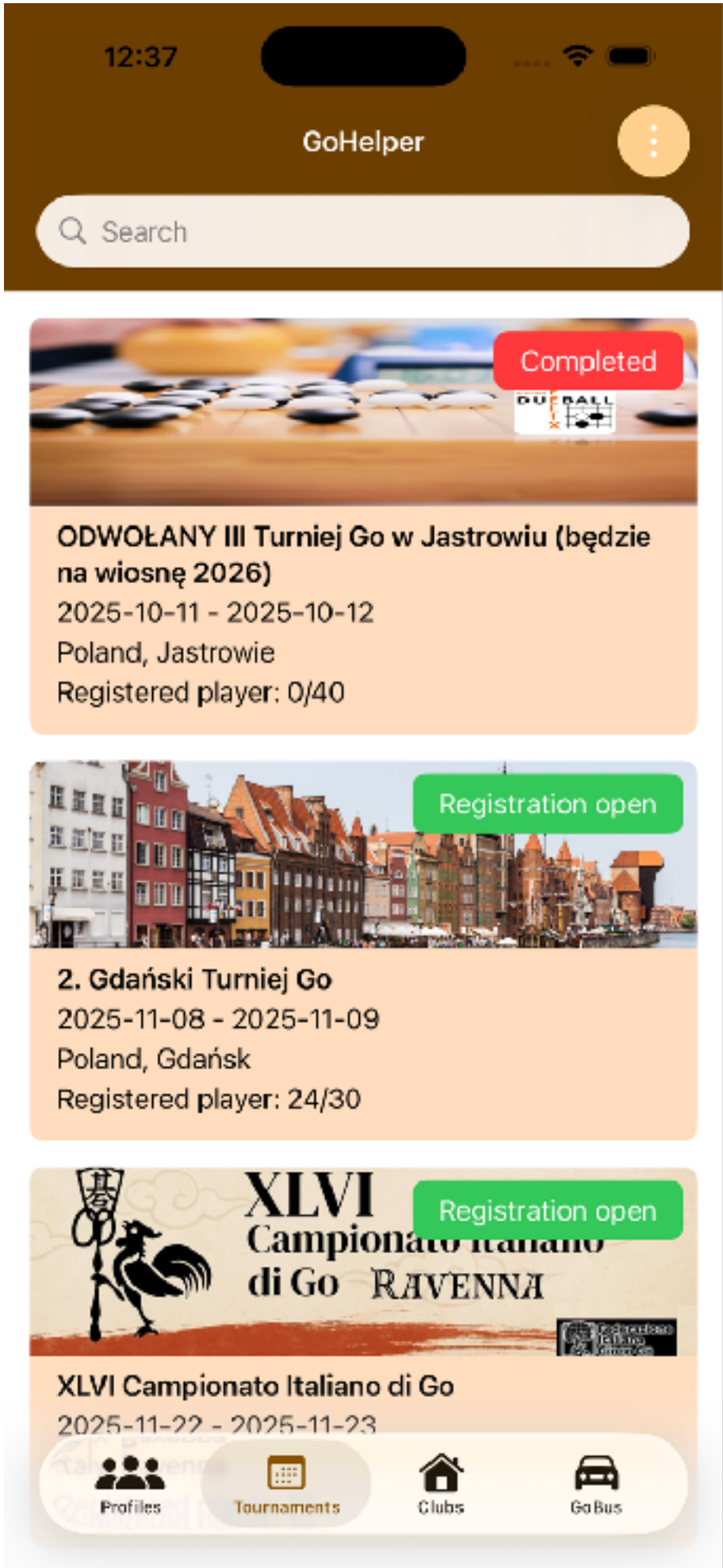
Diagonal (inches)	Models
5.4"	iPhone 12 mini, iPhone 13 mini
5.8"	iPhone 11 Pro
6.1"	iPhone 11, 12, 12 Pro, 13, 13 Pro, 14, 15, 15 Pro, 16, 16e
6.3"	iPhone 16 Pro, iPhone 17, iPhone 17 Pro
6.5"	iPhone 11 Pro Max, iPhone Air
6.7"	iPhone 12 Pro Max, 13 Pro Max, 14 Plus, 14 Pro Max, 15 Plus, 15 Pro Max, 16 Plus
6.9"	iPhone 16 Pro Max, iPhone 17 Pro Max

iPhone models compatible with iOS 26 (3)

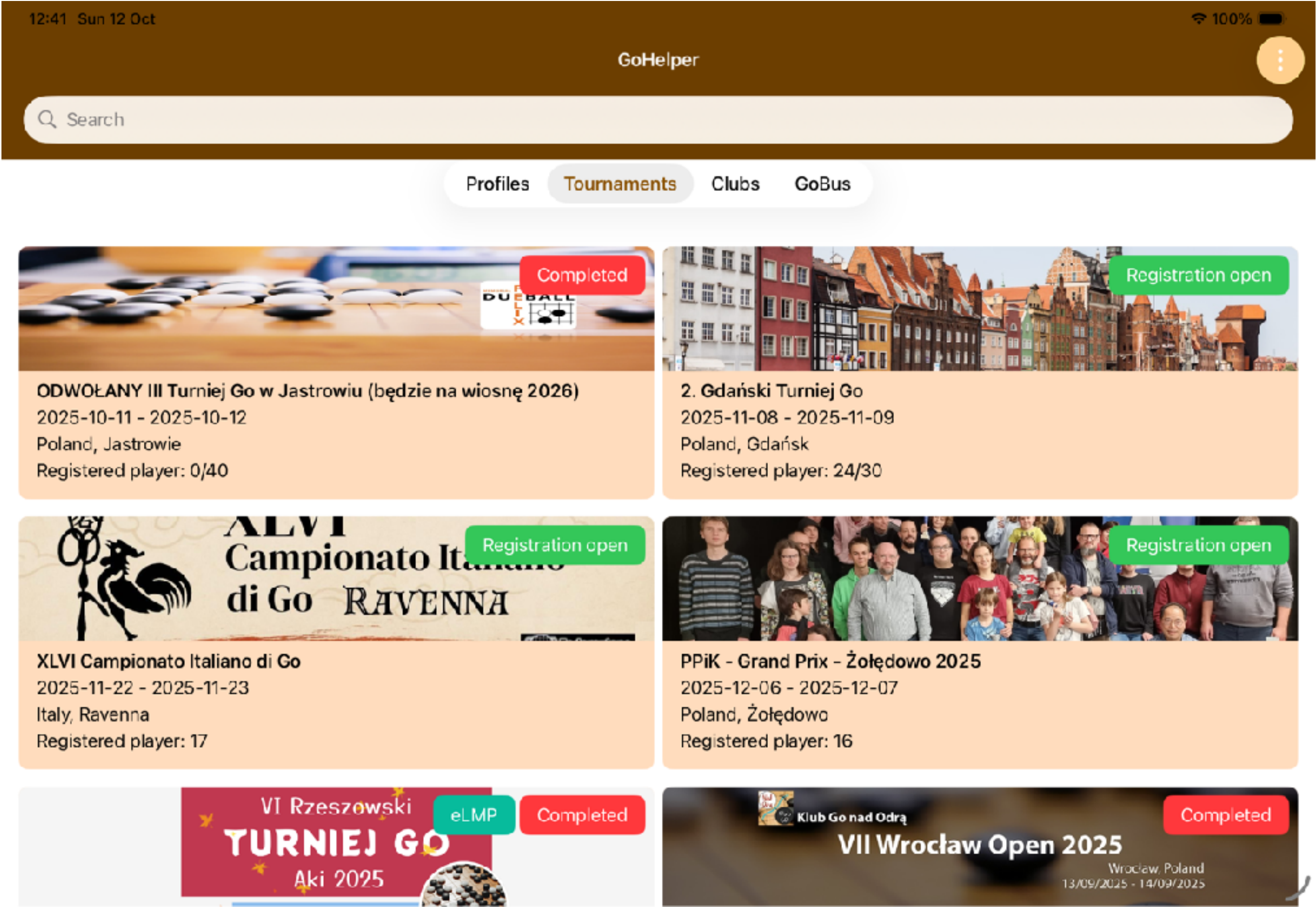


But this is not all we need to consider when
designing UI/UX

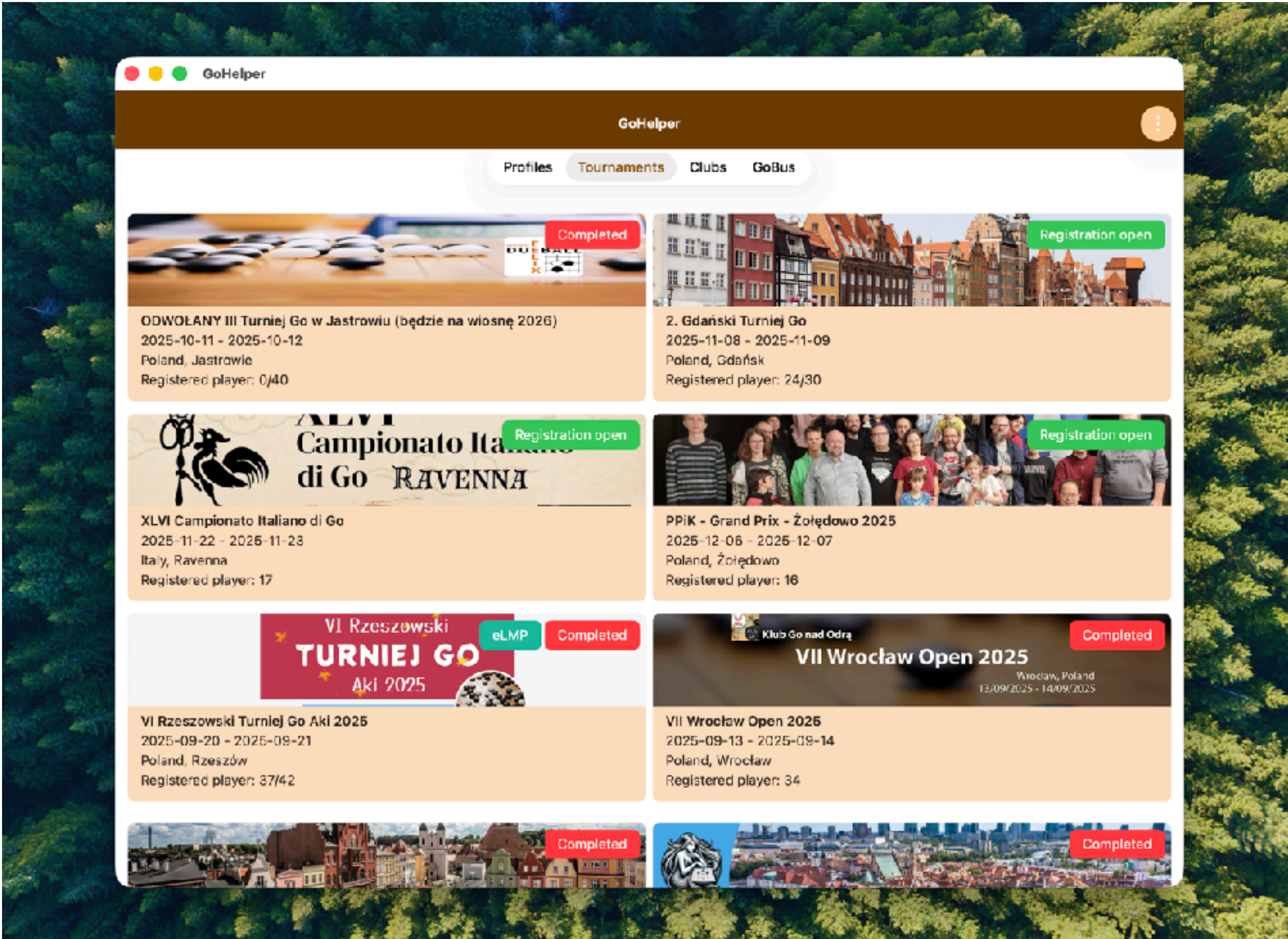
iOS



iPadOS



macOS



visionOS



UI - User Interface

What is UI?

- UI (User Interface) refers to the **visual and interactive elements** of a product.
- It defines **how a product looks and feels**.
- Focuses on **design, aesthetics, and layout** to facilitate interaction.
- Example: Buttons, icons, menus, color schemes.

Key Elements of UI

- **Visual elements:** Typography, colors, icons, imagery.
- **Interactive components:** Buttons, sliders, input fields.
- **Layout & hierarchy:** Guides users' attention and understanding.
- **Animations & microinteractions:** Feedback and engagement.

Role of a UI Designer

- Translate UX wireframes into **visual designs**.
- Create a **cohesive design language** for the product.
- Collaborate closely with UX designers to balance **functionality and style**.
- Ensure visual design aligns with branding guidelines.

UX - User Experience

What is UX?

- UX (User Experience) refers to the **overall experience of a user** interacting with a product.
- Focuses on **usability, satisfaction, and problem-solving**.
- Concerned with **how a user feels and navigates** through the product.
- Example: The flow of booking a flight or purchasing a product online.

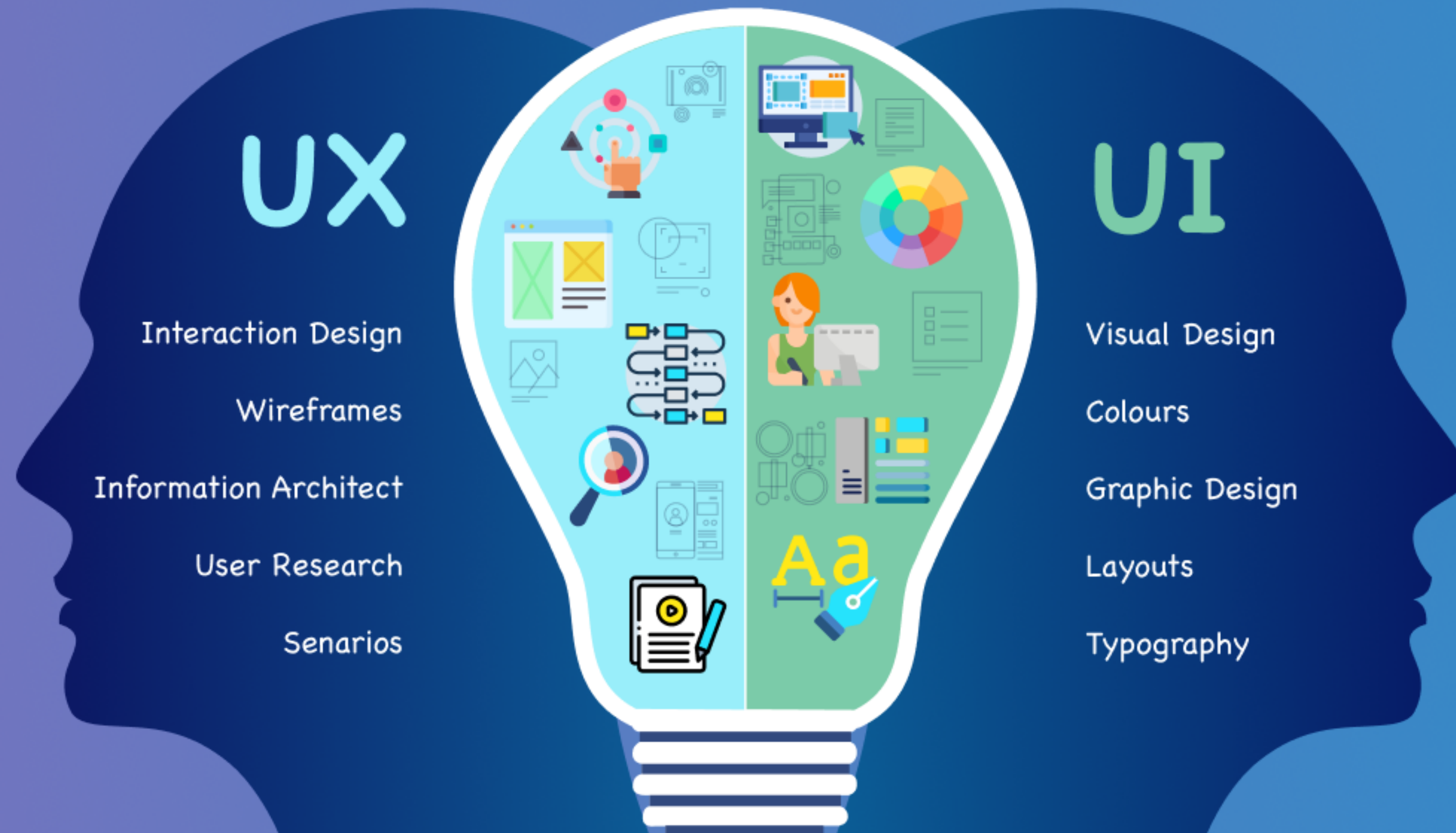
Key Elements of UX

- **User research:** Understanding user needs, behaviors, and pain points.
- **User journeys & workflows:** Mapping the user's path through the system.
- **Prototyping & testing:** Iterative design and usability tests.
- **Information architecture:** Structuring content for ease of use.

Role of a UX Designer

- Conducts **research and analysis** to understand user problems.
- Designs **user flows, wireframes, and prototypes**.
- Validates ideas through **testing and feedback**.
- Ensures **functionality and usability** align with product goals.

The Difference between



Human Interface Guidelines

Human Interface Guidelines

Developer

News

Discover

Design

Develop

Distribute

Support

Account

Design

Overview

What's New

Get Started

Guidelines

Resources

Filter

> Getting started

> Foundations

> Patterns

> Components

> Inputs

> Technologies

Human Interface Guidelines

The HIG contains guidance and best practices that can help you design a great experience for any Apple platform.

As you design interfaces for Apple platforms, keep these principles in mind:



Hierarchy

Establish a clear visual hierarchy where controls and interface elements elevate and distinguish the content beneath them.



Harmony

Align with the concentric design of the hardware and software to create harmony between interface elements, system experiences, and devices.



Consistency

Adopt platform conventions to maintain a consistent design that continuously adapts across window sizes and displays.

For developer guidance, see [Adopting Liquid Glass](#).

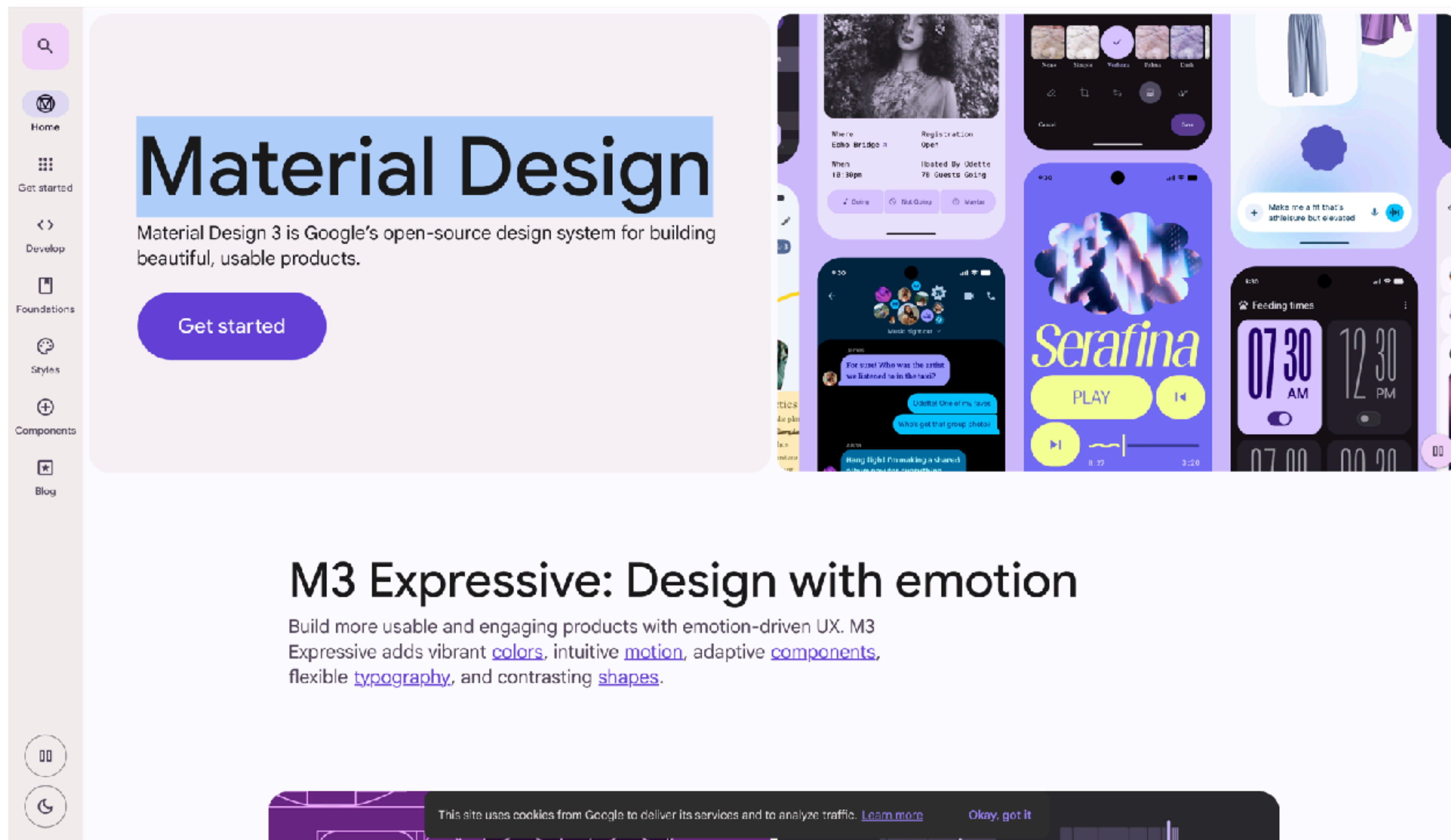
Design fundamentals



Source: <https://developer.apple.com/design/human-interface-guidelines>

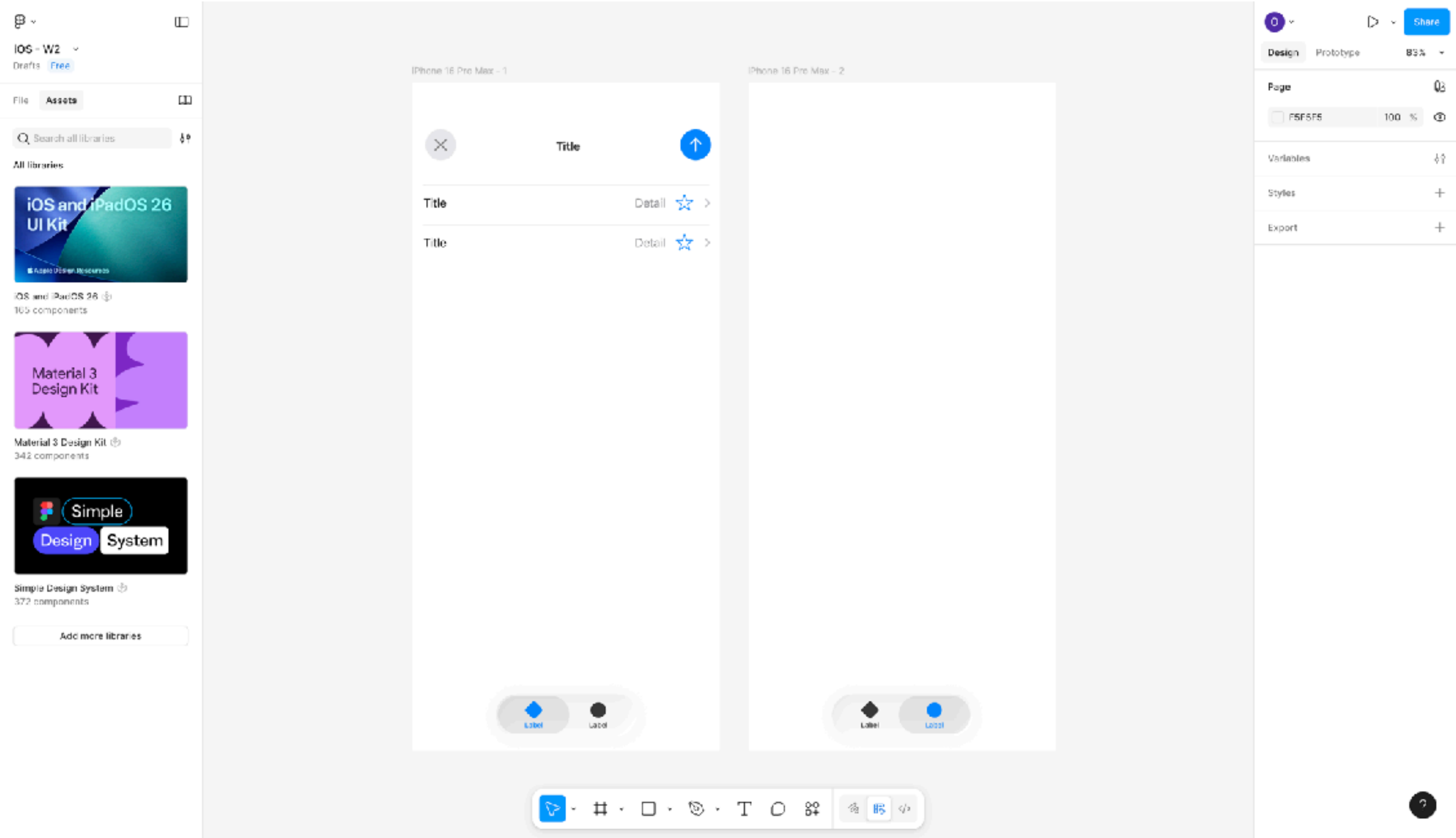
Material Design

Material Design 3



Figma (Demo)

Figma (Demo)

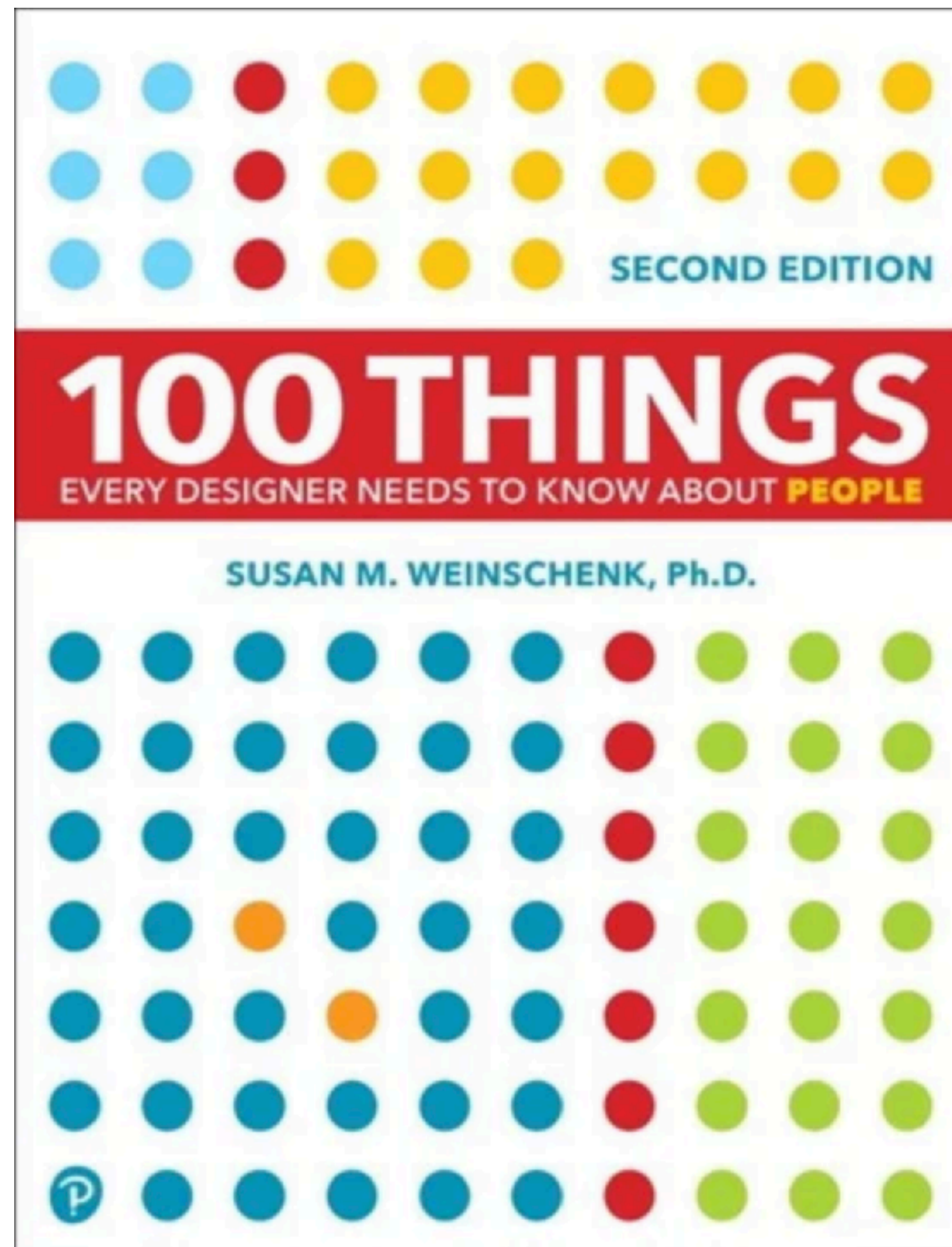




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Books





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Useful links

- **iOS Human Interface Guidelines:** <https://developer.apple.com/design/human-interface-guidelines/designing-for-ios>
 - **Material Design:** <https://m3.material.io/>
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“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
